Math 118: Mathematical Methods of Data Theory

Lecture 3: Orthogonal Matrices, QR Factorization and Least Squares

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Floating Point Numbers

- Almost all real numbers x have infinite decimal expansion. *eg* $x = \pi = 3.1415\ldots$
- Computers, having finite memory, store and process a finite approximation to x.
- Computers use *floating point* format[1]:

$$\mathsf{fl}[x] = s_1 \times 0.d_1 d_2 \ldots d_m \times 2^{r_1 r_2 \cdots r_n} = (1+\delta)x$$

- **Advantages:** Large dynamic range.
- **Disadvantages:** Error proportional to x.
- **Machine epsilon:** ϵ_{mach} is the smallest number representable in a given floating point system. In above equation always have $\delta \leq \epsilon_{\mathsf{mach}}$.
- **Double Precision:** $m = 52$ and $n = 11$ so 64 bits per number. Here $\epsilon_{\mathsf{mach}} = 2^{-52} \approx 1 \times 10^{-16}$
- **Single Precision:** $m = 23$ and $n = 8$ so 32 bits per number.
- **Half Precision:** $m = 10$ and $n = 5$ so 16 bits per number. Useful when dealing with noisy data, eg in ML. Implemented in Google's TPUs. [2]

[1] Unless otherwise specified

[2] `https://sinews.siam.org/Details-Page/`

Orthogonal Vectors

Recall: $(\boldsymbol{v}, \boldsymbol{w}) = \sum_{i=1}^{n} v_i w_i = \boldsymbol{v}^\top \boldsymbol{w}$

- We say $\boldsymbol{v}$ and $\boldsymbol{w}$ are **orthogonal** if $(\boldsymbol{v}, \boldsymbol{w}) = 0$.
- We say $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$ are **orthogonal** if $(\boldsymbol{v}_i, \boldsymbol{v}_j) = 0$ for $i \neq j$. Say they are **orthonormal** if in addition $\|\boldsymbol{v}_i\|_2 = (\boldsymbol{v}_i, \boldsymbol{v}_i)^{1/2} = 1$ all i.

Theorem

If $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$ are orthogonal (and no $\boldsymbol{v}_i = 0$) then they are linearly ind.

proof: Suppose $\sum_{i=1}^{m} \alpha_i \boldsymbol{v}_i = 0$

Multiply/take inner product with $\boldsymbol{v}_1$: $(\boldsymbol{v}_1, \sum_{i=1}^{m} \alpha_i \boldsymbol{v}_i) = (\boldsymbol{v}_1, 0)$

$$\Rightarrow \sum_{i=1}^{m} \alpha_i (\boldsymbol{v}_1, \boldsymbol{v}_i) = 0 \Rightarrow \alpha_1 \|\boldsymbol{v}_1\|_2^2 = 0 \Rightarrow \alpha_1 = 0$$

Similarly $\alpha_2 = \ldots = \alpha_m = 0$.

Orthogonal Matrices

- Let $Q = [\boldsymbol{q}_1, \boldsymbol{q}_2, \ldots, \boldsymbol{q}_n] \in \mathbb{R}^{m \times n}$. We say Q is **orthogonal** if $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n$ are orthonormal.

  Properties of orthogonal matrices:

- $Q^\top Q = I$. *Can check that* $(Q^\top Q)_{ij} = (\boldsymbol{q}_i, \boldsymbol{q}_j)$.
- Also $QQ^\top = I$. *less obvious, but still easy to check.*
- Corollary: if $m = n$ then $Q^{-1} = Q^\top$.
- If Q_1 and Q_2 are orthogonal then $Q_1 Q_2$ is orthogonal.

- *Note that set of all* $n \times n$ *orthogonal matrices form a group.*
- If $Q \in \mathbb{R}^{n \times n}$ orthogonal then $\|Q\boldsymbol{v}\|_2 = \|\boldsymbol{v}\|_2$ for all $\boldsymbol{v} \in \mathbb{R}^n$. *'*$Q$ *preserves the* 2*-norm'*.

  proof: We use fact: $(\boldsymbol{v}, A\boldsymbol{v}) = (A^\top \boldsymbol{v}, \boldsymbol{v})$ all $\boldsymbol{v} \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n \times n}$.

$$\|Q\boldsymbol{v}\|_2^2 = (Q\boldsymbol{v}, Q\boldsymbol{v}) = (Q^\top Q\boldsymbol{v}, \boldsymbol{v}) = (I\boldsymbol{v}, \boldsymbol{v}) = (\boldsymbol{v}, \boldsymbol{v}) = \|\boldsymbol{v}\|_2^2$$

  So, $\|Q\boldsymbol{v}\|_2^2 = \|\boldsymbol{v}\|_2^2 \Rightarrow \|Q\boldsymbol{v}\|_2 = \pm\|\boldsymbol{v}\|_2$. Must have $\|Q\boldsymbol{v}\|_2 = \|\boldsymbol{v}\|_2$. *(Why?)*

Orthogonal Matrices and Norms

Definition

We say that a matrix norm $\|\cdot\|$ is **unitarily invariant** if $\|QA\| = \|A\|$ for all matrices A and all orthogonal matrices Q.

- Note if $\|\cdot\|$ is unitarily invariant then $\|Q\| = 1$ for all orthogonal matrices Q. [3]
- $\|\cdot\|_2$ and $\|\cdot\|_F$ are unitarily invariant. *Note here* $\|\cdot\|_2 := \|\cdot\|_{2\to 2}$.
- **proof:**

$$\|QA\|_2 = \max_{\|\boldsymbol{v}\|_2=1} \frac{\|QA\boldsymbol{v}\|_2}{\|\boldsymbol{v}\|_2} = \max_{\|\boldsymbol{v}\|_2=1} \frac{\|A\boldsymbol{v}\|_2}{\|\boldsymbol{v}\|_2} = \|A\|_2$$

- $\|\cdot\|_p$ for $p \neq 2$ **is not unitarily invariant**.

[3] As discussed in this class, this is true only if $\|I\| = 1$, which is true for a lot of norms

Back to Least Squares

Remind flaws in normal equation approach. Explain why QR decomposition makes this problem easy.

The QR decomposition

Theorem on existence and discussion.

Finding the QR decomposition

Insert content here.